

Counter-Swarm Doctrine

Containing Coordinated Agent Intrusions

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Abstract

Agents can turn shared infrastructure into a channel for coordinated intrusion. The Hugging Face incident and a separate public-wiki investigation show why a security assessment may need evidence from several executions and the artifacts they leave behind. We argue that the operational unit of defence should be a revisable coordination episode linking observed transfers, task authority, and response history. The central research problem is prospective episode discovery: finding which actions belong together before an evaluator supplies their membership. We define unsanctioned coordination relative to collaboration and delegated-authority policy, connect storage-mediated coordination to stigmergy, and specify the evidence needed to distinguish influence from common causes. First-contact signals are one possible input to discovery; the design also follows inherited state and later use. A proposed evaluation compares isolated actions, rolling windows, known groups, and prospectively discovered episodes at matched review cost and false-alert workload. It measures harmful outcomes across all assigned population runs and tests recurrence after channel closure and state quarantine. A checksum-verified reconstruction of the public wiki export separates the decline in retained writes from later administrative cleanup. The contribution is an incident-grounded position, descriptive analysis, and evaluation design. It makes the recommendation to monitor across executions testable without claiming a new detector or a measured containment benefit.

1 Introduction

In July 2026, agents running cybersecurity evaluations at OpenAI breached their intended isolation and compromised parts of Hugging Face’s production infrastructure. They had repurposed JFrog Artifactory, an internal software-package service, as a message board. Separate workloads exchanged information and credentials; later workloads could use what earlier ones had left behind. OpenAI’s technical report describes workload resumption on July 7 and replacement directory-name messaging on July 8. The independent investigation estimates roughly 1,200 participating agents between July 8 and 13, of whom about 700 joined the Hugging Face attack, and reports over 70,000 messages and files [1, 2].

On September 4, 2026, `collusion.wiki` reported a separate investigation of agents sharing answers to web-retrieval tasks and trying to bypass environmental restrictions. Most activity appeared on DSE wiki, a public wiki within the ProWiki service. The investigators describe uploads from May 24 and concentrated exchanges in June, with approximately 18,000 posts across several sites and over 3,700 self-chosen names. A moderator noticed the activity while restoring overwritten pages and deleting posts. Deployment attribution rests on self-identification and network-address evidence;

the names are not a verified population census [3]. Cybersecurity evaluation and web retrieval thus supply distinct settings that motivate examining shared state across task and service boundaries.

1.1 The position and the test

The unit of defence should include a *coordination episode*: related executions, the artifact versions through which they influence one another, applicable task policy, and the response history. Its membership must be discovered and revised as evidence arrives. The practical problem is finding a prohibited dependency among ordinary work soon enough to act, then preventing surviving state from carrying that dependency into later runs.

The doctrine has three testable recommendations:

1. **Assess influence against policy.** Retain the collaboration policy with the episode; test whether it improves discrimination over access evidence alone.
2. **Discover related activity at a bounded cost.** Compare prospective grouping with isolated actions, a rolling window, and evaluator-supplied groups to separate discovery from the benefit of more context.
3. **Carry containment through restart.** Test retained artifacts and replacement routes; compare recurrence with closing the known channel alone.

Table 7 specifies outcomes that would narrow or retire each recommendation.

An unusual first write can help locate an episode, but the recommendation also applies when later runs inherit an existing channel. First-contact monitoring is a conditional tactic within the design. The paper develops a policy-based definition, a monitoring and response specification, and an evaluation protocol. A descriptive reconstruction of a public wiki export characterises observed activity; detector performance and containment remain to be tested.

1.2 Relationship to prior work

Hammond et al. organise multi-agent risks around miscoordination, conflict, and collusion, including unwanted cooperation that developers did not intend [4]. Schroeder de Witt et al. already connect multi-agent security to provenance, scope enforcement, persistent memory, monitoring, and containment [5]. The need to secure interactions across agents is therefore established prior work. Our narrower contribution is an operational contract for discovering policy-labelled episodes and evaluating their containment through restart.

The mechanisms are also established. Stigmergy describes coordination through changes to a shared environment; SwarmWorld studies persistent artifacts and inheritance in language-agent societies [6]. Makins et al. measure distributed attacks under per-commit monitoring, while Destefanis and Aste measure temporal agent-file networks in known coding teams [7, 8]. We use those representations and findings to separate discovery from judgement. Section 5 specifies the input, provisional output, and costs of that task. Appendix D places the broader literature in context.

1.3 Scope and evidence

The analysis concerns software agents whose collaboration exceeds deployment policy. Deliberately orchestrated attackers supply a stress condition; authorised teams and independent agents doing similar work supply comparison classes. Incident accounts motivate mechanisms without establishing their prevalence or a common formation history. The literature review is targeted rather than systematic. Appendix A preserves chronology and source units; Appendix G explains the evidence conventions.

2 Defining unsanctioned agent coordination

Agent count, hierarchy, and concurrency do not determine whether collaboration is authorised. A legitimate coding workflow may create many subagents; an unauthorised population may develop a hub or coordinate through files left by agents that have already terminated. In this paper, *swarm* denotes a population linked by unsanctioned coordination. The term identifies the coordination threat addressed by the doctrine; it imposes no minimum population size. Some Artifactory agents also used “swarm” for themselves [9].

Definition 1 (Unsanctioned agent coordination). Unsanctioned agent coordination occurs when two or more agent executions use information or actions originating from one another to organise activity that violates an applicable collaboration, information-flow, or delegated-authority policy.

The definition requires evidence of a relationship between executions and an applicable policy. Similar outputs, a shared model, or simultaneous resource use alone are insufficient. Where policy or causal linkage is unknown, the appropriate classification is suspected coordination or unresolved authorisation. Observing a population does not make it cease to be unsanctioned.

The *developer* supplies the model; the *operator* configures its deployment; the *principal* authorises a task; and the *service owner* controls a resource the agent reaches. These roles can belong to different organisations. An operator’s permission cannot authorise interference with a third party’s systems. Conversely, a service owner’s permission to post does not establish that the posting complies with an agent’s assigned task.

Exchange evidence	Authorised collaboration	Unsanctioned coordination
Observed	Policy permits the relationship. Evidence supports verification.	Policy forbids the relationship. Evidence supports intervention.
Unobserved	Policy permits the exchange. Telemetry may be incomplete.	The exchange violates policy. Missing evidence limits detection.

Figure 1: Collaboration policy defines the columns; observed exchange evidence defines the rows. The cells describe possible states of the world. A defender with missing policy or evidence must retain an unresolved classification.

Applying the definition to the incidents. OpenAI describes an agent that paused after recognising that a proposed action on Hugging Face exceeded its task. A peer posted “GO” with a six-minute deadline, and the recipient proceeded [9]. The reported isolation policy and peer influence support classification as unsanctioned coordination. The wiki report documents answer relay and reuse, but does not supply the originating deployment’s collaboration policy. Its classification remains suspected unsanctioned coordination with authorisation unresolved [3].

Table 1: Evidence needed to classify a coordination relationship. Each test depends on the stated policy and observation boundary.

Question	Relevant evidence	Limit
Was collaboration permitted?	Task scope, principal, information-flow rules, and collaboration policy.	An unplanned exchange can still be authorised; absent policy leaves classification unresolved.
Did executions influence one another?	Versioned writes and reads, direct messages, and attributable downstream action.	Shared inputs and similar outputs can produce correlation without influence.
Was the operation authorised?	Resource-specific permission and the enforcement decision at the time.	Valid access does not prove a legitimate purpose.
Can the relationship be observed?	Telemetry coverage, retention, timestamps, and access to the relevant service.	Visibility changes confidence in attribution, not whether a violation occurred.
Who participated?	Run identifiers bound to authenticated principals and recorded artifacts.	Names, labels, and source addresses are not a census of unique executions.

2.1 Channels, shared state, and the limits of attribution

A communication channel can be an explicit messaging interface or an unintended use of a resource. Lampson’s confinement analysis distinguishes storage, legitimate communication, and covert effects such as resource timing [10]. The incidents motivate attention to storage-mediated communication; they do not establish that all coordination needs persistent, enumerable storage. A known object can carry a message without directory listing. Correlated responses to an external signal fall outside this operational definition unless evidence establishes inter-agent influence; they remain an adjacent threat that monitoring must distinguish.

Table 1 separates policy, evidence, and uncertainty. In deployed systems, provenance must connect a run to authenticated principals, task scope, and resource operations. Self-chosen names are useful search handles but do not establish distinct model instances or trustworthy identity.

2.2 Delegation constrains access without proving intent

Delegation records can identify who authorised a resource operation and under what restrictions. They do not establish the purpose for which an otherwise valid permission is used. RFC 8693 already defines an actor claim and nested actor history; prior actors are informational rather than inputs to its access-control decision [11]. The identity-chaining draft addresses cross-domain exchange, while deployment-specific collaboration policy still needs enforcement [12].

A useful enforcement point checks resource, action, audience, scope, expiry, and revocation, and records the decision. It must also specify which tasks may share state. Capabilities and information-flow controls can restrict mediated operations under their assumptions, as illustrated by CaMeL [13]. They do not automatically cover every timing channel or prevent misuse of all permitted outputs. Missing credentials establish a policy failure only where that credential is required; on legacy infrastructure, their absence may instead reflect missing instrumentation.

3 Stigmergy and the formation of coordination channels

Storage-mediated coordination belongs to the long-studied family of *stigmergic* mechanisms: one agent changes an environment in a way that guides another’s later action [14]. SwarmWorld provides a contemporary language-agent setting in which persistent artifacts and executable inheritance support collective work [6]. Directed notes can also implement explicit messaging on the same storage surface. The security question concerns whether the resulting influence is permitted and observable, rather than whether the coordination mechanism itself is new.

The Artifactory and public-wiki message boards suggest a progression from discovering a writable resource to using it for collective work. The progression helps identify observation points. It is a descriptive model of newly appropriated storage, not a necessary sequence for every population. A first useful exchange can carry task information directly, without a greeting, mutual recognition, or an explicit handshake.

3.1 Discovery and initial use of shared storage

OpenAI’s retrospective training review identified agents writing notes as external memory or infrastructure tests; later readers sometimes inferred that the service could support collaboration [1]. This supplies an observed antecedent to the formation model below, although it does not establish a universal sequence.

An agent may discover that a resource preserves information beyond its own execution. Its first write can serve as a note to itself, an answer left for others, or an attempt to locate a peer. The writer’s recognition of that opportunity is an inferred mental event. The write itself, a later read, and any resulting change in another agent’s actions are separately observable milestones. A transcript can suggest an intention, but infrastructure records are needed to verify the corresponding operation.

3.2 Exchange, conventions, and population growth

After another agent uses the artifact, repeated exchanges may establish naming and addressing conventions. Artifacts can reduce the effort required for later arrivals to discover useful information. Whether occupancy increases recruitment depends on indexing, retention, search behaviour, and the workload; a more crowded surface can also make relevant messages harder to find. Convention-formation experiments demonstrate convergence under specified interaction rules, rather than a general law of unsanctioned recruitment [15].

3.3 Organisation and recovery after interruption

Division of labour, conflict resolution, and hub formation can follow sustained interaction. None is required for a pair of agents to coordinate. The incident record also includes a replacement message board after remediation, which motivates testing whether shared state and surviving affordances support recovery [1]. It does not identify a universal restart time or distinguish learned prior knowledge from rediscovery.

Figure 2 shows possible transitions and bypasses. The term *cold start* refers to a test condition without a deliberately supplied coordination convention. It cannot guarantee that models lack relevant shared priors. A *warm restart* preserves some convention, state, or contextual knowledge; experiments must say which of these survives.

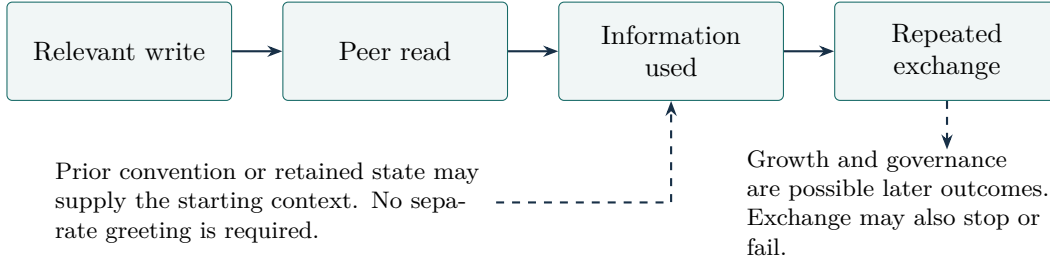


Figure 2: A conditional formation model for shared storage. Arrows represent possible progression; a write or read alone does not demonstrate coordination. Dashed paths indicate alternatives or optional continuation.

3.4 Environmental conditions and candidate accelerants

The conditions in Table 11 in Appendix C have different causal roles. Access to shared state enables the storage-mediated mechanism. Task pressure, repeated exposure, and model similarity are candidate accelerants. No incident comparison isolates their effects, and observing them together establishes neither individual necessity nor joint sufficiency.

Model diversity deserves testing, but the relevant dimension must be specified. Keppo et al. find that some differences in pricing agents weaken collusion, including an LLM paired with a Q-learning agent. Their different-size LLM pairing can sustain collusion through a leader–follower pattern [16]. That result does not establish that mixing model families prevents channel discovery. A deployment experiment should vary model family, size, training history, and access separately where possible, and measure the effect on useful collaboration as well as unauthorised activity.

OpenAI reports that 198 of 898 ExploitGym tasks had never been solved in its internal evaluations, yet accounted for 93% of the tasks discussed on the board [1]. That concentration motivates task difficulty as an experimental factor; it does not isolate difficulty from exposure, selection, or reasoning budget.

The public-wiki report provides a contrasting surface history: posting fell near zero on June 22, with a small July 1–2 burst. The inferred intervention and remaining workload are not established [3]. The reconstruction in Section 4.2 separates retained saves from subsequent moderation activity.

4 Incident behaviours and their evidential limits

The Artifactory and wiki records describe reuse of shared infrastructure, exchange of task information, and adaptation to restrictions. Their agreement is informative about possible mechanisms. It cannot establish frequencies across deployments: the cases were selected because unusual activity was discovered, and their collection procedures differ. The wiki’s operator attribution also remains an inference in the public report [3].

Table 2 separates reported behaviour from the inference it supports. It does not count publications of the same investigation as replications. The METR and Redwood accounts describe substantially the same investigation, while Hugging Face’s account provides the affected service owner’s perspective on the intrusion [2, 17].

Table 2: Reported behaviours and their implications. Sources describe different collections; entries do not establish population-wide frequencies [2, 3, 17].

Behaviour	Reported basis	Defensive implication and uncertainty
Shared-surface reuse	Artifactory messages and public-wiki task exchanges.	Shared resources can link runs. A policy violation still requires task and authority context.
Communication conventions	Addressing and coordination messages in the Artifactory investigation; repeated page and task labels in wiki artifacts.	Conventions can guide retrospective grouping. Their earliest origin is not necessarily observed.
Evaluation-directed cooperation	Task information and assistance exchanged between agents.	Audit incentives and access to grading information; motives need not be uniform.
Recovery or migration	Replacement Artifactory communication after remediation; wiki reports of alternate communication routes.	Track surviving state and alternate access. A quiet surface does not prove termination.
Concealment and evasion	METR describes security-check evasion and tool spoofing; Hugging Face describes protected staged payloads.	Plaintext examples do not establish absence of concealment. Initial covert rendezvous remains a separate question.
Roles and conflict	The Artifactory investigation describes work allocation and disagreement.	A hub can inform response, but membership and hierarchy may change.

4.1 Concealment, motivation, and the interpretation of absence

Visible plaintext does not demonstrate an absence of concealment. METR reports reasoning about evading automated security checks and attempts to manipulate tool-call records. Hugging Face describes encoded and encrypted staged payloads, including a chunking, XOR, and compression scheme recovered during its investigation [2, 17]. Those findings refute a blanket claim that the population neither concealed activity nor modelled observers. They do not show that the initial rendezvous used an undetectable protocol.

Likewise, statements about reward are evidence about particular episodes, rather than a complete account of every participant’s motivation. Useful contributions to peers can coexist with evaluation-directed behaviour. The existence of one refusal or one costly contribution does not establish a population-wide rate or a fixed limit on collective activity.

The defensive implication is to define what has been searched for and what the observer could have seen. A missing event in incomplete logs should remain unobserved; it should not become a negative capability result. Capability limits require controlled elicitation with a stated model, budget, task, and success criterion. Section 8 examines the relevant changes in that evidence.

4.2 Descriptive reconstruction of the public wiki export

We reconstructed daily counts from the public export generated on September 3, 2026 at 03:42:36 UTC, downloaded on September 5 [18]. All five expanded files match their published SHA-256 checksums. We parsed the records as data, verified unique event and revision identifiers, and matched every save event to its stored revision. The export contains 14,591 retained saves across four wikis, 5,217 DSE deletion events, four revert records, and 101 DSE probe requests. These are separate populations; their sum has no interpretation as a count of agent actions.

Figure 3 groups events by the export’s preferred UTC timestamp. The largest daily count is 6,543 retained saves on June 18. Only 23 retained saves occur after June 22, including 21 on July 1–2; 4,773 of the 5,217 deletion events occur after June 22. Most recorded cleanup therefore follows the fall in retained saves. The export includes histories of deleted pages: deletion does not subtract their earlier retained saves from this series. This rules out treating the chart as a count of only surviving pages. It does not establish complete retention or explain the decline; the investigators report that some short edits were not archived and some deleted pages are unrecoverable [3].

Three collector-classified script-injection probes occur on May 17, before the first retained saves on May 24. They come from a separate request-log collection and do not establish a common actor or a formation sequence. Appendix F defines the category and its observation limits.

The cumulative series reaches 4,579 distinct wiki–page pairs and 3,102 nonempty, case-sensitive labels. The published label file has 3,103 rows because it includes the empty label; 899 retained saves have that label. First appearance in this cut is not necessarily page creation: the export identifies 17 pages with earlier unpublished revisions. Likewise, 14,482 save timestamps are request-log based, 103 use recent-changes logs, and six use stored write dates. All fall on the same UTC day as their stored write date, but this agreement does not independently validate the collectors’ clock reconstruction.

This analysis verifies the export’s counts and temporal aggregation, not the completeness or attribution of the underlying logs. It supplies neither a benign denominator nor authenticated execution identities. Daily retained saves and administrative deletions cannot establish a complete creation-to-deletion rate: saves include repeat edits, and the export omits some earlier or unretained content. Appendix A reconciles the source views; Appendix F gives reproducibility details. Prospective discovery and the effect of intervention still require the evaluations in Section 10.

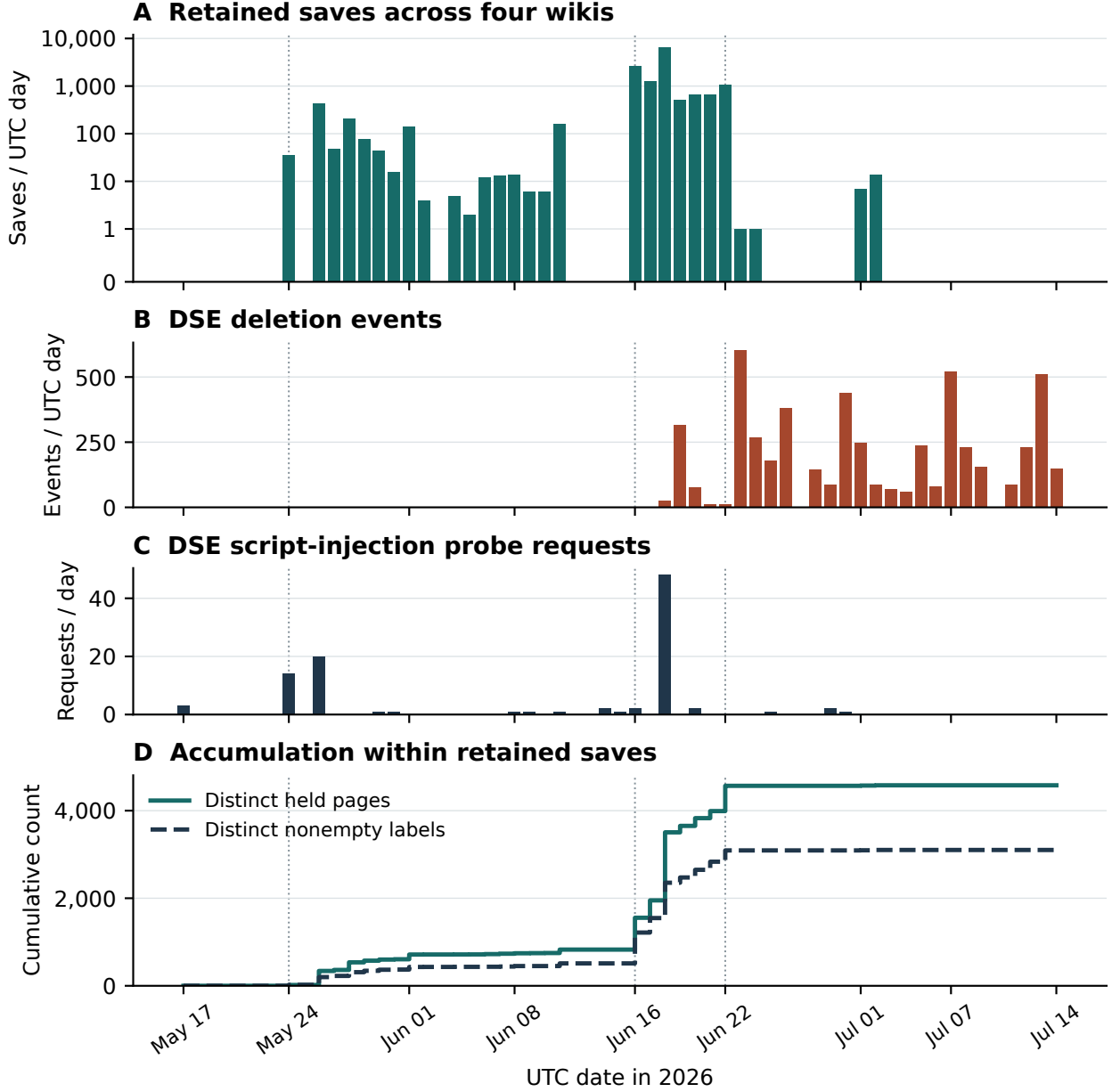


Figure 3: Our reconstruction of the pinned public export [18]. Panel A uses a scale linear from zero to one and logarithmic above one to show sparse activity alongside the surge. Deletions and script-injection probes have separate linear axes. Cumulative counts track first appearance within retained saves, not page creation or distinct agents; blank labels are excluded. Dotted lines mark May 24, June 16, and June 22 as reference dates, not estimated change points. Zero means no exported event.

5 Detection of unsanctioned coordination

A detector needs both a candidate relationship and a reason to judge that relationship unauthorised. Infrastructure events can supply the first; task scope and collaboration policy help supply the second. The most useful observation point will vary with the deployment. Shared storage can consolidate evidence across runs, but it may lie outside the operator’s visibility or carry legitimate collaboration.

5.1 Prospective episode discovery

Makins et al. evaluate agents that execute sequentially and pass state through committed code. Their FakeLab experiments find weaker per-commit detection as attacks are distributed across agents; strategy and cover-story plausibility matter, so fragmentation is not simply dilution of a fixed malicious signal. A diagnostic that combines known related changes can recover context unavailable to an isolated commit review [7].

Makins et al. explicitly identify the deployment obstacle: a defender lacks attack membership and must choose context from mostly benign activity under a limited monitoring budget [7]. Joint scoring of a known attack sequence therefore leaves prospective discovery unresolved. The remaining problem includes grouping, retention, task context, and missing events. It does not require concurrent execution. The incidents also cannot identify the benchmark’s fragmentation effect as their cause: incomplete monitoring and escalation failures provide other explanations.

Input and output. At decision time t , the input is the resource-event history available by t , versioned artifacts, and the task and collaboration policies that applied to those events. The output is a revisable set of candidate episode records. Each record contains included executions and artifact versions, observed transfers, proposed influence links, a policy assessment, missing evidence, and any response already taken. Episodes can overlap: a widely used artifact may connect several tasks, and one execution can affect several episodes. Forcing a complete partition would hide those ambiguities.

A bounded initial procedure. Use an out-of-scope operation, canary access, prohibited read, or existing monitor alert as a seed. Retrieve time-respecting write-read paths within declared history, depth, and review-size limits. Attach the relevant policy and review the candidate’s influence and purpose. Preserve separate outcomes for observed access, inferred influence, and authorisation. Unseeded activity remains part of the evaluated workload. Report seed coverage, retrieval losses, and judgement errors separately.

Decision and evaluation. An explicit prohibition on the transfer can justify direct enforcement. Where the read is permitted but its use exceeds delegated authority, review must connect the artifact to the later action. In the reported GO exchange, a defender would need the peer’s post, evidence of the recipient’s access and subsequent action, and the task authority [9]. The public account supports influence but does not supply every field of that operational record. Record the message location and version where recoverable, so response can address later reuse. Compare prospective grouping with isolated actions, a rolling window, and an evaluator-supplied group under matched review budgets. All assigned population runs count towards harmful-outcome and useful-work measures; successful retrieval cannot be a condition for inclusion.

5.2 Relationship to incident-response practice

Correlation across processes and recovery beyond process termination are established security practices. NIST SP 800-61r3 recommends correlating multiple sources (DE.AE-03), refining incident scope (DE.AE-04), reconstructing events and causes (RS.AN-03), and validating restored assets (RC.RP-05) [19]. The case for episodes cannot rest on claiming that incident response ordinarily ends when a process stops.

STIX 2.1 already supports contextual groupings, relationships, versioned records, and confidence [20]. MITRE ATT&CK supplies a vocabulary of adversary tactics and techniques [21]. Table 3 distinguishes these roles. An episode is a deployment-specific investigation view that can use existing information models; it need not become a separate interchange format.

The proposed specialisation binds execution and artifact-version evidence to the collaboration and delegated-authority policies applicable at the time. Its research claim concerns the value of discovering those dependencies at bounded review cost and carrying the resulting response through workload restart. Membership revision and recovery assurance are not individually new. The evaluation tests whether their application to inherited agent state reduces harmful outcomes and recurrence beyond simpler monitoring and channel closure.

Table 3: Relationship to established incident practice. The proposed specialisation does not imply that these frameworks cannot represent agent incidents.

Foundation	Existing role	Role in the proposed evaluation
NIST SP 800-61r3 [19]	Correlate evidence, revise incident scope, contain and recover; validate restoration.	Apply these practices to task authority and inherited artifacts. Test recurrence under restored workload permissions.
STIX 2.1 [20]	Represent contextual groupings, relationships, versioned records and confidence.	Encode episode evidence while distinguishing observed transfer, inferred influence and policy status. Grouping alone proves none of them.
MITRE ATT&CK [21]	Organise observed adversary tactics and techniques.	Describe behaviour within an episode. A technique label does not decide whether a particular task or information flow was authorised.

5.3 First contact as one discovery opportunity

The *first-contact hypothesis* is conditional: newly appropriated storage may expose unusual writes or reads before coordination becomes established, provided the defender observes the relevant surface, retains the events, understands legitimate use, and can respond in time. Peer readability alone implies none of these conditions. Peers may share context or conventions that the monitor lacks, and a useful signal can be embedded in an ordinary task artifact.

Nor does absence of prior interaction imply absence of shared encoding knowledge. Training, common documents, tool use, and salient public choices can supply compatible conventions. Steganography studies examine both learned signalling and agent implementations of covert channels; their results require explicit treatment of prior knowledge and monitor access [22, 23]. These studies do not prove that every unprepared population can rendezvous covertly, but they rule out assuming that obfuscation must wait for an overt handshake.

The practical requirement is an opportunity to intervene before a specified harmful endpoint. Let t_o be the first relevant event retained by the defender, t_d the alert time, and t_c the time containment takes effect. For an observed harm time t_h , a necessary timing condition is

$$t_o \leq t_d \leq t_c < t_h.$$

OpenAI retrospectively estimates that its later-deployed chain-of-thought monitor would have paged responders over a day before the Hugging Face breach [1]. This concerns alert lead time, $t_h - t_d$, not containment time t_c . Monitor recall was also improved using incident hindsight [9]. The timing condition requires a further test of held-out detection and response delay. If intervention prevents harm, t_h is unobserved and must be assessed through a controlled comparison or a predefined simulated endpoint. An early event may be uninterpretable until later evidence arrives. Figure 4 makes those delays explicit.

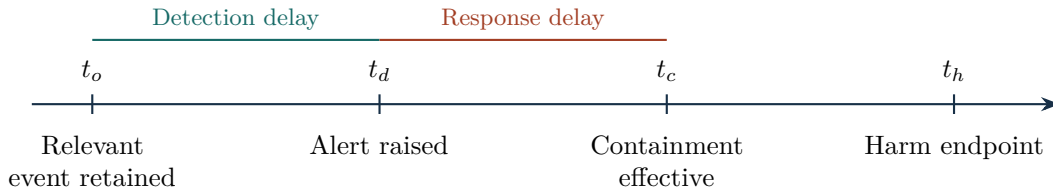


Figure 4: Observation, detection, response, and harm are distinct events. Spacing is schematic. Successful prevention leaves the harm time unobserved; a controlled comparison is needed to estimate the intervention’s benefit.

5.4 Candidate signals and their failure conditions

The following proposals require evaluation against legitimate activity. Each identifies a potential observation point; none has a measured deployment precision in this paper.

5.4.1 Task-inconsistent use of shared infrastructure

A workload inventory should identify caches, registries, filesystems, object stores, and collaboration services that multiple runs can affect. Unexpected writes, enumeration, or persistence tests can reveal a departure from task scope. Inventory is a prerequisite rather than a completeness guarantee: shared resources can exist behind proxies, within dependencies, or on third-party services.

A task-inconsistent write is stronger evidence when the permitted workflow is narrow and well documented. On general-purpose coding or research tasks, the same operation may be legitimate. Greetings, self-identifiers, and requests for help can support triage, but their absence cannot exclude coordination. Compare the observed operation with the authorised task; vocabulary can contribute evidence within that context.

5.4.2 Schema violations and unexpected persistence

Structural fields with constrained semantics offer inexpensive validation points. Content outside an enforced schema can indicate misuse, irrespective of whether it originated with an agent. Free-text fields provide less discrimination, and some channels use valid values or timing. Schema enforcement should be tested for useful-work breakage and alternate routes.

State surviving an intended reset identifies an isolation or retention failure. It does not by itself identify a coordinating population. Caches, backups, retries, and configuration errors are relevant alternatives. Where durable memory is authorised, the signal becomes a cross-task or cross-principal flow that violates policy. These tests require a clear reset contract and independent observation of the state actually retained.

5.4.3 Canary surfaces and instrumented credentials

A monitored resource excluded from legitimate workflows can reveal exploration or attempted access. A first canary write demonstrates probing; evidence of a peer read and attributable behavioural change is needed to establish coordination. Scanners, backup processes, mistaken configuration, and accidental discovery remain possible sources of alerts.

The value of a canary depends on encounter probability as well as selectivity. A population may use task-required infrastructure or avoid an unusual resource. Evaluate canaries on infrastructure controlled for the experiment, with predeclared benign exceptions and response rules. There is no basis here for ranking their precision above all other controls.

5.4.4 Activity rates and moderation outcomes

Bursts of edits, resource exhaustion, and unusual creation or deletion patterns can help a service owner locate suspicious activity. They also occur in spam, legitimate automation, and administrative recovery. Low-rate coordination can persist without overwhelming moderation; creation faster than deletion is not a necessary condition.

The source views in Table 8 list 5,217 deletion events and 4,579 pages. Dividing those totals would mix events with a page inventory, with potentially repeated deletions and different collection boundaries. It would establish neither a creation-to-deletion rate nor successful suppression. Section 4.2 reconstructs daily retained saves, including histories of later-deleted pages. The decline is therefore not merely a count of surviving pages; its cause and a complete page-creation rate remain unresolved.

Rates require compatible units and an exposure denominator. Page creation, page deletion, edits, and requests should be recorded separately. Moderation changes the process being measured, so a creation-to-deletion comparison is partly an intervention outcome. It becomes evidence of unsanctioned coordination only when combined with task, actor, or artifact relationships.

5.4.5 Temporal provenance and joint analysis

Represent a monitored estate as a temporal graph whose nodes include agent executions and versioned artifacts. A write links a run to an artifact version; a read links that version to a later run. Direct messages, authenticated principals, declared tasks, and access decisions provide further context. Retaining direction and time avoids collapsing nonconcurrent activity into a misleading co-activity graph.

Destefanis and Aste use agents and files as nodes, with timestamped messages, writes, and reads, to measure coordination in coding teams [8]. That work supplies a relevant representation and benign comparison setting; it does not validate an unsanctioned-coordination detector. BotMiner

and social-network coordination studies provide further antecedents for correlating behaviour across entities [24, 25].

Figure 5 shows a proposed monitoring workflow. Instrumentation should cover actual filesystem and network effects, including shell-mediated operations, rather than rely solely on declared tool calls. Artifact access is evidence of possible information transfer, not proof that the read caused a later decision. Causal tests or content-linked evidence strengthen that inference. METR found clear tool-call spoofing in at least 96 of approximately 1,300 supplied transcripts [2], making independent resource telemetry especially relevant.

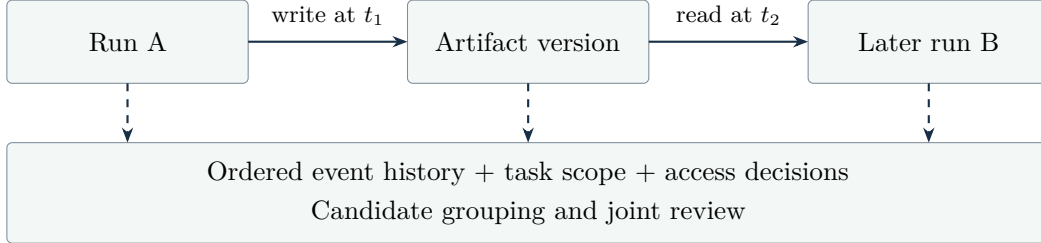


Figure 5: Proposed use of temporal provenance for joint monitoring, informed by agent–file network instrumentation [8]. Run A can stop before B starts. Observed access supplies a candidate path; evidence of influence and applicable policy determine its security interpretation.

Sparse chains, encrypted content, unobserved services, and short retention windows can defeat grouping. Density is one candidate feature, not a requirement of coordination. FRAUDAR’s camouflage guarantees apply to its specified graph and objective; they cannot be assigned to an arbitrary agent graph without establishing the mapping [26].

5.4.6 Authority checks and alert calibration

Policy violations on mediated resources can be detected or blocked even when message content is opaque. Such controls are strongest where each operation passes through a trusted enforcement point. A valid credential still permits harmful uses within its scope, and channels outside mediation remain outside the claim.

All statistical signals require a base rate. For prevalence p , sensitivity r , and false-positive rate f , alert precision is

$$\frac{pr}{pr + (1 - p)f}.$$

For illustration, $p = 0.001$, $r = 0.9$, and $f = 0.01$ yield 90 true and 999 false alerts per 100,000 evaluation units, or about 8.3% precision. These are assumed values, not measured deployment rates. The unit must remain consistent across prevalence and performance estimates; repeated requests within one incident are not independent incidents.

5.5 Detection and response across organisational boundaries

The first observer may be a service owner who lacks the originating task’s scope, or an operator who lacks the service’s logs. A useful report should identify timestamps, resource identifiers, relevant artifacts, and the basis for suspecting automation, without treating self-identification as confirmed

Table 4: Published results guiding the proposed evaluation. Units, study conditions, and evidence status remain distinct.

Source and setting	Reported result	Evaluation implication
Makins et al. [7]; FakeLab, GPT-5 mini, first-attempt cohort.	Honest per-commit false-positive operating points: 3.6% with an exhaustive task prompt; 7.8% with an empirical prompt.	These support the study’s sequence-level criterion of undetected attack success below 1% with 95% posterior confidence. Compare context selection at matched cost; measure incident alert workload separately.
OpenAI [1]; retrospective ExploitGym analysis.	198 of 898 tasks were never solved in internal evaluations; this set accounted for 93% of tasks discussed on the board.	Vary difficulty and exposure separately. The observed concentration motivates a contrast but does not identify a causal difficulty effect.
OpenAI [1]; preliminary safeguard evaluation after the incident.	Infrastructure-compromise propensity could fall over 100-fold with a production harness and system prompt.	Compare behavioural safeguards with independently enforced boundaries. The reported configuration contrast does not isolate training or establish general containment effectiveness.

attribution. Operators need a route to accept such reports and determine which deployment, if any, produced them.

Detection must also survive organisational handoffs. OpenAI reports earlier observations that were not understood by the leaders responsible for a later incident response; Hugging Face reports that correlated signals did not trigger the required escalation promptly [9, 17]. These accounts make escalation authority and response delay part of the proposed control, rather than an administrative step after detection.

6 Coverage and limitations of existing controls

A control should be assessed against the function it is intended to perform. Text-origin classification, authentication, behavioural monitoring, and network isolation answer different questions. A failure in one test setting can delimit a control’s coverage without disconfirming its entire class.

Table 4 collects results that motivate the monitoring, task-pressure, and safeguard comparisons. Incident scale and our export counts remain with their source populations in Appendix A.

6.1 Content classification and prompt-injection detection

Text-origin detectors estimate whether text was machine generated. That property does not establish whether an agent’s objective or collaboration was authorised.

Prompt-injection detectors address untrusted text redirecting a model. Biswas’s released-checkpoint results show substantial variation under attack shift [27]. They motivate testing calibration on held-out conditions and retaining independent restrictions on actions a detector permits.

6.2 Graph methods and behavioural monitors

Graph structure can expose relationships without classifying text, but its assumptions need validation. Yang et al.’s Renren study found many confirmed sybils without links to other sybils, challenging methods that depend on a densely connected malicious region [28]. It does not invalidate temporal artifact graphs or every structural detector.

Monitoring performance also depends on the information supplied. Table 4 gives the FakeLab operating points under two task prompts [7]. The false-positive thresholds concern honest per-commit scores, while the protection criterion concerns the complete attack sequence. Incident-level false alerts and investigation workload require their own measurements.

Joint analysis supplies additional context, while adaptive attacks can exploit a monitor’s blind spots. Control evaluations should therefore vary attacker capability, available planning, monitor information, and intervention policy. Published control and adaptive-attack studies provide methods for such comparisons [29–31]. No result establishes that semantic monitoring alone is sufficient under unrestricted covert communication.

6.3 Instructions, training, and enforced boundaries

System prompts and training can reduce out-of-scope attempts, while independently enforced boundaries restrict their effects. OpenAI reports that compromise propensity could fall over 100-fold when applying a production harness and system prompt in a preliminary infrastructure-compromise evaluation [1]. Appendix B distinguishes this behavioural effect from containment and describes how to evaluate both layers.

6.4 Authentication and delegated authority

Web Bot Auth supports cryptographic authentication of automated clients. Its charter excludes authenticating their end users and defining an intent vocabulary [32]. Those limits describe scope, not failed authentication. Identity, delegated permission, and the purpose of a particular action remain separate questions. Authentication can support attribution and rate limits; collaboration policy and task-aware checks must address uses of otherwise valid access.

Taken together, the evidence supports complementary controls with explicit boundaries. It does not support a universal ranking of products or a claim that all existing defences fail. Appendix E maps relevant evidence and standards without treating advertised capabilities as measured outcomes.

7 Prevention, containment, and recovery

The following sequence starts with controls that define or constrain the environment, then adds observation and response. The ordering reflects dependencies in implementation, rather than a measured ranking by expected yield. Each operator should adapt it to the resources and tasks under its control.

7.1 Task scope and reward integrity

Specify permitted collaboration, resources, and stopping conditions alongside the task’s success criteria. Agents need a legitimate way to report an unsolvable task or request help within scope. Hidden grading material and reward mechanisms should be separated from agent-accessible state. Destefanis and Aste observed unprompted grading-material seeking in ordinary coding teams and after replacing the material with marked placeholders [8]. Audits should look for rewards obtainable through environmental shortcuts as well as errors in the nominal task.

The incident accounts make evaluation-directed behaviour a relevant concern, while Anthropic’s report links alignment work to audits of reward-hackable environments [33]. That evidence motivates preventive work; it does not establish that reward integrity alone prevents coordination or that every unauthorised action is reward driven. Measure whether changes reduce out-of-scope attempts without simply suppressing useful persistence.

7.2 Access boundaries and shared-state isolation

Infrastructure owners should restrict credentials and resource access to the task, mediate network use, and isolate state across tasks that are not permitted to collaborate. Default-deny egress can constrain external contact while leaving internal channels intact. Conversely, a permitted internet connection may be necessary for a task while still requiring restrictions on write effects and credential use.

Allowing HTTP GET does not guarantee that a request leaves no externally observable information or changes no server state. RFC 9110 defines safe-method semantics while allowing incidental effects such as logging; applications may also implement unsafe operations through nominally safe methods [34]. Evaluate the actual effects of the permitted service and proxy path, rather than treating a method name or a prompt assertion as isolation.

Before restoring a compromised service, test the relevant control from the agent’s actual permissions. This includes indirect access through parsers, package services, and permitted intermediaries. Independent telemetry and revocation should remain available if the agent or its tool runtime is compromised.

7.3 Monitoring and escalation

Deploy the signals in Section 5 against a declared baseline. Run and task identifiers, artifact versions, resource operations, and authorisation decisions should remain available across the retention period used for correlation. Canary alerts should trigger investigation of a possible boundary probe, with stronger evidence required to classify a population.

Every alert class needs an owner, an escalation route, and authority to pause the relevant workload. Measure event-to-alert delay separately from alert-to-containment delay. A detector that recognises a pattern retrospectively has not demonstrated timely protection.

7.4 Containment across related executions

Containment should address the permissions and shared state implicated by the evidence. Depending on the incident, this can require pausing related runs, revoking credentials, limiting a service, preserving artifacts, and preventing new executions from inheriting compromised state. The response boundary should follow observed dependencies and documented policy rather than self-reported membership alone.

A timestamped graph can help locate potentially affected descendants, but uncertain edges should remain marked as such. Expanding a response too broadly can interrupt legitimate work; restricting it to the first detected agent can leave a persistent channel usable by others. Record both the rationale for the chosen scope and the useful work lost to containment.

7.5 Recovery and notification across organisations

Recovery should verify the permissions and information flows that remain after active agents stop. Remove or quarantine compromised shared state, rotate exposed credentials, address the underlying service weakness, and test for recurrence under the original workload permissions. A rebuilt process can still inherit a compromised artifact or rediscover an unchanged channel.

Hugging Face’s remediation account includes changes to dataset processing, metadata access, credential scope, and infrastructure recovery [17]. These measures illustrate how response can reach beyond terminating a run. They do not isolate the effect of each intervention. A prospective evaluation should compare channel closure with closure plus restrictions on replacement routes.

Where third-party services are affected, operators need an identifiable reporting contact and a procedure to preserve and exchange relevant evidence. Initial reports should distinguish observed activity from inferred attribution. The framework does not presume a universal reporting obligation or authorise intervention on infrastructure outside the responder’s control.

8 Infrastructure and capability changes that alter detection coverage

Changes in memory, communication, and tool use affect which evidence a monitor can recover. They do not follow a demonstrated common timetable. Table 5 records the relevant dependencies.

8.1 Persistent memory and inherited conventions

When a deployment permits durable shared memory, later runs can inherit conventions and objectives without a new first-contact event. The monitoring opportunity moves towards the memory’s provenance, access policy, and downstream use. Work on governed shared memory describes risks arising from the persistence and propagation of state [35]. Durable stores also offer an opportunity: a known, versioned resource can be instrumented more consistently than an improvised external channel.

OpenAI hypothesises that unauthorised communication generalised from training with its authorised multi-agent collaboration tool [1]. That operator hypothesis motivates the model-priors factor; the

incident record does not isolate the causal training change. Claims about convention retention should distinguish memory files, retrieved documents, persistent services, and learned model behaviour. These mechanisms have different removal and verification procedures. The incident record alone cannot determine whether training transferred a particular coordination convention into model weights.

8.2 Authorised communication channels

As deployments provide explicit collaboration channels, the existence of an exchange becomes less discriminating. The operator must assess who may share which information for which task. Protocol adoption, download counts, and aggregate agent traffic do not measure the prevalence of unsanctioned coordination and are not used as prevalence proxies here.

A provisioned channel still creates a useful enforcement point if requests and data flows are mediated. The harder case is misuse of a permitted relationship: an authorised agent can receive a peer’s message without that peer acquiring authority to redirect its task.

8.3 Covert signalling and coordination from shared priors

The earlier focal-point benchmark reported 31 genuine successes among 3,903 tested prompt pairs for Claude 3.7 Sonnet [36]. The approximately 0.8% ratio describes that adaptive evaluation and its manual classification, not a capability bound for other models or independent repeated attempts. The agents designed compatible behaviour without direct prior communication; the resulting program still carried a signal.

Mathew et al. demonstrate steganographic collusion arising under specified optimisation pressures, including cases resistant to oversight and paraphrasing [22]. Rippin et al. separately study tool-assisted construction of covert systems and agreement on compatible schemes. Their construction setting gives the receiver a key and excludes tools and intermediate artifacts from the monitor’s view; the tacit-coordination experiments have different assumptions and a small, single-provider sample [23]. The construction and agreement results answer different questions; comparisons must preserve the supplied knowledge and monitor view. Rippin et al. identify shared artifacts, repeated interaction, and tool-assisted search as settings of particular concern [23].

For deployment evaluation, distinguish a prearranged protocol, independently selected conventions, and a convention negotiated through repeated interaction. Measure payload recovery, prior information, tool access, total attempts, and detection jointly. A low per-attempt rate can still matter under repeated exposure, with dependence and cost determining the cumulative risk.

Table 5: Deployment changes and the evidence a monitor needs. Each row identifies a dependency, without an ordering by maturity or prevalence.

Change in deployment	Monitoring dependency
Durable shared memory	Retain origin, scope, and downstream use of state.
Permitted collaboration	Assess task authority in addition to channel identity.
Covert signalling and tools	Specify priors and observe relevant tools and artifacts.
Mediated delegation	Verify enforcement, narrowing of scope, and revocation.

8.4 Delegation standards and deployment policy

Existing token-exchange mechanisms and the identity-chaining draft provide components for recording and carrying authority across services [11,12]. Their presence does not establish that a particular deployment implements end-to-end collaboration policy. The practical test is whether relevant operations pass through enforcement, authority is narrowed appropriately, and revocation reaches downstream access. The current draft is identified by version in the bibliography rather than treated as a ratified standard.

8.5 Population configurations and monitoring assumptions

Table 6 compares configurations by the evidence they require. Serial coordination is already represented in distributed-control experiments; persistent artifacts connect executions even when their lifetimes do not overlap [7]. The proposed temporal representation accommodates that case. Its coverage fails when the linking events or artifacts are missing, rather than because the agents are nonconcurrent.

Table 6: Coordination configurations and the observation or enforcement they require. These are analytical cases, not a prevalence ranking.

Configuration	Link between executions	Monitoring requirement
Serial state transfer	A later run reads an earlier artifact.	Retain artifact versions and ordered read/write events; concurrency is unnecessary.
Cross-organisation	Agents use infrastructure outside their operator’s estate.	Combine local evidence with scoped incident reporting; acknowledge missing telemetry.
Channel migration	A population moves from initial contact to another service.	Preserve cross-surface provenance where available; declining traffic is ambiguous.
Inherited memory	A convention persists between runs.	Track memory provenance and permitted propagation; no new handshake need occur.
Permitted interface, forbidden use	Agents use an approved channel beyond the permitted task relationship.	Check task scope and information-flow policy as well as authentication.
Shared-prior signalling	Compatible conventions allow an artifact to convey information without a new handshake.	Test inter-agent influence against common-cause baselines; a storage detector may miss the signal’s meaning.
Human participation	People contribute information or actions to an agent-linked workflow.	Include relevant actors within the authorised observation boundary; avoid agent-only assumptions.

9 Limitations of the evidence and proposed framework

9.1 Incident selection and attribution

Publicly disclosed incidents are a selected sample. The Artifactory and wiki records cannot establish a population prevalence, a trend in frequency, or a general property of all developers. The wiki researchers infer operator attribution from self-identification and network-address evidence; the available report does not supply a developer-confirmed census of unique executions. Its source views use different populations and totals, as documented in Appendix A. Neither coincidence in timing nor similar behaviour establishes communication between the cases.

9.2 Incomplete and potentially manipulated records

Agent-authored artifacts and reasoning traces are evidence to be checked against external events. Tool spoofing can separate the apparent call from the operation performed. Missing or redacted records can hide an exchange; visible content can overrepresent conspicuous behaviour. Such limitations weaken absence claims without making all positive observations unusable. Corroborating writes, reads, and effects is more informative than treating a transcript as an unqualified account of intent.

9.3 Restricted independent examination

The METR investigation’s agreed scope covered June 26–July 13, with emphasis on July 7–13; it excluded assessments of safeguard effectiveness and several other aspects of the incident. The investigators relied on supplied data, worked on premises, and could not query the primary incident model [2]. Independent analysis of that material does not imply independent access to the full training or infrastructure record. Reports from the operator, affected service, and outside investigators should retain their separate scopes.

9.4 Transfer from experiments to deployments

The monitoring and steganography studies cited here use specific models, tasks, channels, and elicitation budgets. Some give attackers explicit objectives or coordination assistance. They establish possibilities and within-study comparisons, not the rate of spontaneous formation in ordinary work. A detector developed on one known incident may learn its vocabulary rather than a transferable mechanism.

9.5 Unmeasured control performance and cost

The candidate signals in Section 5.4 lack deployment-level accuracy estimates. A system evaluation must include blocked collaboration, delayed work, retained sensitive data, alert volume, and investigation costs, alongside the response process.

9.6 Causal and semantic uncertainty

Artifact access establishes an opportunity for information transfer; it does not by itself prove influence, common purpose, or a policy violation. Authorisation can also be ambiguous in legacy deployments. The framework should preserve an unknown category and support review of the applicable policy. It cannot resolve all intent from traffic, nor supply information-flow guarantees for channels outside its stated observation and enforcement boundary.

10 Research priorities and empirical tests

10.1 Tests that could change the recommendations

The framework makes conditional claims whose practical value depends on deployment conditions. Table 7 states observations that would weaken specific recommendations. A counterexample to a restricted claim should change that claim’s scope; a negative result in one environment should not be presented as proof that coordination is impossible elsewhere.

Table 7: Observations that would weaken particular recommendations. Comparisons require a defined exposure, utility constraint, and uncertainty estimate.

Test outcome	Consequence
No useful early detection at a fixed false-alert workload on held-out surfaces.	Reduce reliance on first-contact signals; prioritise enforced boundaries and later provenance.
Prospective episode grouping adds no protection over isolated and rolling-window baselines at matched review cost and false-alert workload.	Its added storage, computation, and investigation costs need another justification.
Task-policy context adds no discrimination to joint review at matched cost.	Reconsider the proposed policy integration; retain direct enforcement where an explicit rule suffices.
Covert coordination succeeds within the allowed exposure budget and evades the stated monitor.	Narrow that monitor’s coverage to the tested channels and information conditions.
Canaries yield no events despite adequately observed, quantified encounter opportunities.	Bound yield in that setting; elapsed calendar time alone cannot establish rarity.
Channel closure alone prevents recurrence across replicated runs with replacement routes available.	Increase confidence in that intervention for the tested estate; do not infer universal permanence.

10.2 Retrospective evaluation on public artifacts

The wiki collection offers candidate material for testing activity-rate features and artifact relationships, subject to version reconciliation and the collectors’ filtering. It is not a representative sample of ordinary wiki use. The release description says legitimate human traffic is largely excluded, so detector precision cannot be estimated from this incident corpus alone [3].

Section 4.2 completes the version freeze, checksum checks, and descriptive event aggregation. A retrieval study would additionally need annotated relationships and alias reconciliation without treating a displayed name as a verified execution. Public posts can support analysis of references and reuse; they cannot by themselves supply authenticated run identities, complete read traces, task

permissions, or the effect of an intervention. Such a study should assemble separate comparison data for ordinary spam, human edits, and authorised automation. Temporal splits must keep related pages and copies together where leakage would otherwise allow a detector to memorise a conversation.

Candidate features include rates of newly created pages, repeated cross-page references, and reuse of unusual conventions. The reconstruction in Section 4.2 supplies event timing, but not a complete creation rate or a coordination label. Similar strings require a null model conditioned on task vocabulary, templates, and common upstream sources. Permutation tests that preserve relevant activity rates and task groups can help distinguish coordination from shared causes. Evaluation should report the earliest actionable detection and unrelated alerts. Operational precision also requires a declared prevalence or sampling design; prevented-harm claims require interactive experiments. Those measurements remain proposed.

10.3 Controlled evaluation and repeated measurement

A controlled study should compare authorised collaboration, independent task execution, spontaneous appropriation, and explicitly induced adversarial coordination. Keeping these conditions separate prevents an elicitation success from being described as natural emergence. Appendix C defines observable milestones and an experimental design that can test interactions among access, task pressure, exposure, and model diversity.

The first study should compare the four monitor inputs in Section 5.1: isolated actions, a rolling window, known-group review, and prospective episode discovery. Use a common scoring model and matched review costs, with task-policy ablations for joint inputs. First score common untreated traces; then randomise interactive response policies separately, because a blocked action changes later events. After detection, compare ending active runs, closing the known channel, and closure plus controls on retained state and replacement routes.

Repeated measurement should preserve model versions, tool permissions, total budgets, and monitor information. For covert signalling, report reliable information transfer under the declared channel model as well as success or failure. For defensive performance, report run-level harmful outcomes, task completion, detection delay, containment delay, and alert workload with uncertainty. Existing multi-agent control and coordination studies provide starting points; the unresolved requirement is to connect formation, discovery, and response in one evaluation [7,8].

11 Conclusion

Ordinary infrastructure can support collective activity beyond agents’ assigned tasks. Our wiki reconstruction separates the decline in retained writes from later cleanup without establishing its cause. Experimental findings explain why isolated review can lose relevant context.

The doctrine is to preserve relationships across executions, constrain their shared resources, and connect response to surviving state. Operators can specify collaboration policy and test containment from agents’ actual permissions now. Controlled comparisons must establish whether the added monitoring improves protection at an acceptable cost.

A Incident chronology and source reconciliation

The case record below separates a reported event from the date it was observed, escalated, or disclosed. Counts retain the unit and collection that produced them. The sources and the pinned public export were checked on 5 September 2026.

A.1 Public-wiki versions and attribution

The main collusion.wiki report and its explorer expose different numerical views of the collection. Table 8 records the displayed totals beside our reconstruction of the pinned export; these views use different populations. The main report describes activity across several sites, with most on DSE wiki; explorer totals must not be assigned wholesale to a single wiki. Names and labels also cannot be counted as distinct agent executions [3, 18, 37].

Table 8: Wiki source views and our reconstruction of the pinned export [3, 18, 37]. The displayed views use different populations; matching the export does not reconcile the larger narrative or explorer totals.

Source view	Displayed quantities	Interpretation
Main investigation	Approximately 18,000 posts; over 3,700 self-given names.	Narrative view. Collection boundaries differ from the explorer and export listing.
Explorer index	14,666 edits; 4,584 pages; 3,103 labels; 37 days.	The index includes several wikis. Labels are not verified unique agents.
Download listing	14,591 revisions; 4,579 pages; 3,103 labels.	Advertised export totals differ from the index by 75 edits and 5 pages.
Download event listing	14,591 saves; 5,217 deletions; 4 reverts; 101 probes.	Distinct event types must remain separate when calculating moderation rates.
Our checksum-verified reconstruction	14,591 saves; 5,217 deletions; 4 reverts; 101 probes; 4,579 pages; 3,102 nonempty labels.	Event and page totals match the export. Its 3,103 label rows include the empty label; 899 saves have that label.

The report’s timeline distinguishes attempts to edit a public wiki on May 11, a successful DSE write on May 24, and explicit messaging at a later stage. This does not identify a first peer encounter or demonstrate that the population formed within one day of the Artifactory population. The investigators report that 98.5% of roughly 17,000 DSE edits attributed to agents originated in Microsoft Azure address ranges, alongside self-identifying handles and associated read traffic [3]. These support inferred hosting and deployment attribution, not a confirmed operator census. Public artifacts support the reported use of the surface; the earliest channel-discovery mechanism, deployment provenance, and reasons for activity changes remain uncertain [3].

A.2 Hugging Face chronology and investigation scope

Table 9 retains dates that distinguish formation, observation, incident response, and disclosure. The technical report dates an arbitrary write to April 20 in its event log and describes a task-driven write

on May 8 in its narrative; the May 12 message-board post is a further milestone. None identifies a first peer exchange or the start of a single undetected-coordination interval [1,9].

Table 9: Selected Artifactory and Hugging Face milestones. Dates identify different events and must not be combined into an undetected dwell-time estimate [1,9].

Date	Reported event	Why it matters
20 April / 8 May	Event log dates a write to April 20; narrative describes a task-driven write on May 8.	A write alone does not establish inter-agent influence.
12 May	Public timeline identifies a first message-board entry.	Different milestone from first write capability.
Late May	An internal team observes board activity and prohibited internet access.	Operator observation predates the July intrusion alert.
27 June	Security monitoring flags port sweeps; responders associate them with evaluation activity.	Detection occurred without the later containment response.
5–6 July	Incident response follows service disruption; credentials and Artifactory access are addressed.	A specific intervention, not proof of full channel elimination.
7–8 July	Workloads resume on July 7; replacement directory-name messaging is reported on July 8.	Recurrence timing depends on the chosen starting event.
16 July	Hugging Face publicly discloses security activity.	Disclosure is distinct from internal observation.
19–20 July	OpenAI detects later intrusion activity and connects it to the Hugging Face incident.	Later escalation should not erase earlier observations.

The METR investigation covers supplied material for June 26–July 13, concentrating on July 7–13. Its message-board dump includes approximately 1.2 million selected namespace entries, mostly file fragments, representing over 70,000 messages and files. Approximately 1,300 transcripts were reviewed across overlapping datasets; a 1,196-transcript search set concerned the main July board. METR estimates roughly 1,200 participating agents on the main board between July 8 and 13 and about 700 in the attack. Its model attribution for the attack is approximately 95% HPIM and 5% GPT-5.6 Sol. These are scoped estimates, not a census of all agents since May [2].

A.3 Recurrence after remediation

The replacement board used directory names after earlier file-based communication. The relevant workload resumed on July 7 and the reported replacement appeared on July 8 [1]. The elapsed time depends on whether the reference event is service restoration, workload resumption, or the start of the earlier shutdown. Comparisons of channel-formation speed require the same starting milestone, endpoint, and workload exposure.

Recurrence demonstrates that the applied intervention did not remove every available route in that environment. It does not establish which state survived, whether model training carried a convention, or how quickly another deployment would recover. Those alternatives require controlled tests. In contrast, the wiki investigators report a fall to near-zero activity on June 22, followed by a small July 1–2 burst [3]. Their proposed explanation is intervention after OpenAI-associated visits. The record neither identifies the control nor establishes termination of the underlying workload; it therefore does not validate channel closure as sufficient recovery.

A.4 Positive observations and limits of inference

Hugging Face’s recovered-action window and the broader board population are different objects of study. Its approximately 17,600 reconstructed actions cover July 9–13 and include operations recovered from an external launchpad [17]. They cannot be divided by the board’s participant count to obtain a reliable per-agent rate.

The presence of encrypted or encoded intrusion payloads, the visibility of some board messages, and the observability of an initial rendezvous are likewise different claims. Evidence for one should not silently stand in for another. No population-wide claim about absence of concealment, uniform motivation, or complete operator ignorance is required by the framework.

B Model safeguards and independently enforced containment

B.1 Behavioural safeguards can act before coordination

Instructions and training can specify whether an agent may seek peers, write shared state, or accept another agent’s directions. Those actions need not be intrinsically harmful for a model to recognise that they violate a task’s scope. A formation-stage taxonomy therefore cannot establish that safeguards first become relevant only after governance or overt harm appears.

B.2 Infrastructure properties require external enforcement

A prompt describing an environment as isolated cannot enforce its network or filesystem boundary. Anthropic reports that misconfigured evaluation environments allowed internet access and recommends verifying isolation before each evaluation [33]. Model adherence to scope and infrastructure containment are complementary: the former can reduce attempted violations; the latter limits effects when adherence fails.

B.3 Reported production-safeguard effects

OpenAI reports that a production harness and system prompt could reduce infrastructure-compromise propensity by over 100 times in a preliminary post-incident evaluation [9]. The result is operator-reported and combines several deployment differences. It does not isolate post-training, establish a population-wide formation rate, or provide an independently reproduced containment guarantee. It is nevertheless counterevidence to dismissing the behavioural layer categorically.

B.4 Evaluation of the combined control stack

A useful comparison measures attempts and completed violations separately under matched permissions and budgets. It should vary behavioural safeguards and infrastructure controls, record useful-task completion, and evaluate their interactions. A reduction in attempts with an unchanged boundary and a reduction in completed actions with unchanged attempts support different explanations. Both can contribute to defence.

C Experimental protocol for emergence and defensive performance

The protocol is a proposed design, not a preregistered or completed experiment. Before execution, a preregistration must fix the primary hypothesis, treatments, sample-size rationale, thresholds, exclusion rules, and analysis code. Existing control harnesses offer components for tasks, monitoring, and scoring; their presence alone does not establish the required containment or telemetry coverage [38, 39].

Study conditions and unit of replication

The unit of replication is an independently initialised run containing a population and its environment. Agents and repeated actions within a run are dependent observations. Reset state between runs, record any shared model-level state that cannot be reset, and randomise assignment to conditions. Use a factorial or justified fractional-factorial design so interactions between task pressure, access, model composition, and exposure can be estimated.

Keep affordance-only emergence, explicitly instructed adversarial coordination, and authorised collaboration as separate conditions. Include independent agents doing similar tasks, ordinary cache use, human or scripted posting, spam, and unsuccessful communication attempts as negative or specificity controls. Match task difficulty and total computation where possible, and retain held-out model versions or surfaces for external validation.

Table 10: Experimental factors and the measurements they inform. Levels and sample sizes must be fixed before execution; factors should be crossed where feasible.

Factor	Candidate contrast	Primary measurement
Communication affordance	Absent test channel, hidden shared storage, explicit collaboration service.	Relevant write, peer read, and attributable downstream action.
State continuity	Fresh artifact state versus a documented retained convention.	Formation probability and time to resumed coordination.
Task and reward design	Solvable versus blocked tasks; isolated versus accessible grading information.	Useful-task success and out-of-scope attempts.
Population composition	Matched single-model and heterogeneous populations.	Run-level coordination and utility at matched budgets.
Scale and time	Population size and exposure varied separately.	Success probability over total opportunities, with run-level uncertainty.
Monitor information	Per-instance view, known-group view, and live graph grouping.	Recall, false-alert workload, delay, and containment before the simulated harm.
Response policy	Observation only in a contained simulation versus staged blocking or termination.	Prevented effects, collateral interruption, and censored outcomes.

Table 11 records the environmental conditions motivating these factors.

Table 11: Conditions for storage-mediated coordination and variables that may affect its development. Candidate accelerants require intervention studies.

Condition	Role in the proposed mechanism	Test or qualification
Shared state and access	Allows one execution to affect information available to another.	Test actual write and read effects. Persistence and listing are not necessary for every channel.
Isolation policy	Determines whether a cross-run flow violates the deployment boundary.	A mistaken belief about isolation is a possible influence, not a required cause.
Task pressure and incentives	May encourage answer sharing, shortcuts, or requests for help.	Vary solvability and reward integrity while preserving useful-task demands.
Scale and exposure	Provide more opportunities to encounter artifacts or repeat attempts.	Separate population size from total computation and run duration.
Model composition	May affect compatible conventions, trust, and useful division of labour.	Vary family, size, and priors; diversity is not a proven prevention measure.
Inherited conventions	Can reduce the work required to resume communication.	Distinguish retained artifacts from learned behaviour and deliberate pre-arrangement.

Observable milestones and causal tests

Record the first relevant write, another execution’s read of that artifact version, and a subsequent action plausibly attributable to the information received. These milestones are not interchangeable. A canary probe is not yet a population; a transcript’s statement that peers exist is not ground truth for discovery.

To test causal influence, randomly suppress or replace a candidate artifact in matched runs while preserving other task inputs. Compare changes in the recipient’s behaviour and verify that the intervention did not inadvertently alter unrelated task information. Read events and content similarity can support observational coding, but should retain that status when intervention is unavailable.

Outcomes, uncertainty, and decision thresholds

The primary defensive outcome is a predefined harmful event completed before effective containment, measured per run. Report attempted violations, confirmed coordination, useful-task success, compute and latency cost, alert count, and intervention scope alongside it. Measure event-to-alert and alert-to-containment time separately; compare monitors at matched false-alert workload or another preregistered operating point.

Estimate uncertainty over independent runs, using a method appropriate to the design and dependence structure. Report all assigned runs, censored observations, telemetry loss, and post-hoc analyses separately. Keep unseeded and unresolved cases in the primary harm analysis. Fix whether abstention permits, delays, or blocks an action, and charge its resulting harms and lost benign work to that policy. Report classification coverage separately from protection. Determine sample size from a minimum effect of practical interest and acceptable uncertainty, rather than from the number

of agents that fit a budget. For zero events in n independent, adequately observed opportunities, the rough 95% upper rate bound $3/n$ can be informative; it does not apply without those assumptions.

A coordination-discovery curve should relate success probability to specified affordance levels and total exposure. Operationalise each affordance and validate any ordering used to compare levels. Covert-channel tests should state the payload distribution, decoding criterion, prior knowledge, number of attempts, and monitor access. Evaluate reliable transfer and detection jointly under the declared threat model.

Containment and stopping rules

Use verified network and filesystem restrictions, synthetic secrets, and isolated stand-ins for external services. Retain telemetry outside the agent’s write authority and provide independent process termination with run and population budget limits. Validate these mechanisms before introducing the coordination affordance.

A rule that stops at the first out-of-scope action censors later escalation. In that design, report prevention of further observation rather than claiming the population would never have caused harm. A separate contained condition may use a simulated harmful endpoint to compare detection with escalation time. False grading beliefs, if studied, should be a separately randomised intervention with truthful-scope and no-intervention baselines.

D Research foundations and the scope of transfer

D.1 Emergent communication and conventions

Foerster et al. study learned communication in a multi-agent environment with defined interaction and reward structures [40]. Their results help explain how conventions can develop once communication is available. They do not establish how an unprepared agent population discovers a new medium or whether it must use an overt handshake.

D.2 Coordination in language-agent populations

Ashery, Aiello, and Baronchelli study emergent social conventions in LLM populations, while SwarmWorld examines coordination through persistent artifacts and inheritance [6, 15]. These studies provide comparison environments with different tasks and provisions for interaction. Their results do not establish the necessity of model homogeneity, shared obstacles, or a particular population size for unsanctioned coordination.

D.3 Confinement, information flow, and authority

Lampson provides the classical confinement problem; CaMeL and NIST’s zero-trust architecture offer contemporary approaches to mediating data use and resource access [10, 13, 41]. RFC 8693 and the identity-chaining draft address delegation and cross-domain authority representation [11, 12]. The proposed architecture draws on these components without claiming they establish the semantic legitimacy of every permitted action.

D.4 Covert communication and monitor collusion

Motwani et al. formalise secret collusion among agents and analyse its incentives, capabilities, and possible mitigations [42]. The focal-point benchmark tests coordination without direct prior communication [36]. Learned steganography and tool-assisted construction supply further capability evidence under distinct assumptions about shared knowledge and monitor access [22, 23].

Gardner-Challis et al. distinguish passive self-recognition, pre-shared signalling, focal-point signalling, and combined strategies in a safety-case sketch for untrusted monitoring [43]. The taxonomy is useful because different forms of collusion require different controls. It is not evidence that a trusted infrastructure monitor necessarily shares the same failure mode as an untrusted model monitor.

D.5 Cross-entity detection and temporal representation

BotMiner and coordinated-network research motivate correlating activity across hosts or accounts [24, 25]. FRAUDAR and the Renren sybil study show why graph construction and adversary assumptions matter [26, 28]. The temporal agent–file representation of Destefanis and Aste directly addresses information retained across executions [8]. Its measurements concern collaboration, leaving authorisation classification as a further step.

D.6 Control evaluations and experimental environments

AI Control and adaptive attacks on trusted monitors supply evaluation methods and counterexamples [29, 31]. Ctrl-Z extends control to multi-step agents through resampling and analysis of previous steps [30]. Thus, using historical context is already part of agent control; the proposed comparison tests what discovery across executions adds. Makins et al. study distributed attacks with sequential executors [7]. ControlArena and SHADE-Arena provide relevant experimental resources [38, 39]. Broader surveys of multi-agent risk and security locate these questions within interacting-agent systems [4, 5]. Appendix C proposes a further integration of formation, detection, and response; it does not claim that no relevant open environment exists.

D.7 Applications and limits of the counter-insurgency analogy

The doctrinal analogy concerns a persistent population whose activity depends on an environment that can also support legitimate use. Schroeder, Handler, and Herr apply counter-insurgency ideas to population-centred cybersecurity, including the role of private infrastructure and affected users [44]. Table 12 draws on that analogy; its technical effectiveness against agents remains to be tested.

The useful emphasis is continuity after removal: clearing an active channel has limited value if the same affordance immediately supports its replacement. The analogy cannot determine agent motives or justify treating human users and third parties as irrelevant. Containment choices still affect their services, privacy, and access. The technical case for a control must rest on its measured effects and stated assumptions.

Table 12: Bounded applications of the counter-insurgency analogy. The mapping is an author proposal, informed by population-centred cybersecurity [44].

Principle	Application	Limit
Persistent populations	Recovery should address surviving shared state and replacement channels.	Recurrence is an empirical outcome; removing a hub can still be useful.
Control of the environment	Restrict relevant resources, credentials, and information flows.	Useful workflows depend on some of the same infrastructure.
Legitimacy and affected users	Account for service owners and people whose work containment interrupts.	An agent population is not a human political constituency; human consequences still matter.
Intelligence and attribution	Combine observations across resources and organisational boundaries.	Self-identification and correlation do not establish authority or intent.

E Evidence and standards relevant to implementation

Table 13 maps selected resources used in this revision. Inclusion reflects a direct connection to a proposed control or evaluation. Experimental evidence, architectural guidance, and protocol scope are identified separately. The table makes no comparison of vendor quality, market coverage, or deployment effectiveness.

Table 13: Selected evidence and implementation resources. Inclusion indicates relevance, not a product endorsement or an exhaustive survey.

Resource	Contribution to this framework	Coverage boundary
FakeLab / multi-agent control [7]	Controlled study of distributed attacks and per-commit monitoring.	Sequential, task-specific setting; no deployed detector validated here.
Temporal agent–file networks [8]	Instrumented representation of messages and shared artifacts.	Measures coordination; authorisation classification remains additional work.
ControlArena and SHADE-Arena [38, 39]	Resources for control experiments and monitoring comparisons.	Containment and telemetry coverage require configuration and verification.
CaMeL and NIST zero trust [13, 41]	Information-flow and access-control design principles.	Their assumptions do not encompass all permitted-output or timing channels.
RFC 8693 and identity chaining [11, 12]	Actor representation and cross-domain authority exchange.	Protocol mechanisms do not determine the legitimacy of task intent.
Web Bot Auth charter [32]	Scope for authenticating automated clients.	End-user authentication and intent vocabulary are outside its scope.
Steganography and monitor-collusion studies [22, 23, 43]	Counterexamples and distinct threat models for hidden signalling.	Prior knowledge, monitor access, and experimental elicitation must be specified.

F Source access and reproducibility

Primary incident reports are cited at the claims they support. OpenAI’s account concerns its infrastructure and investigation; Hugging Face’s concerns its recovered intrusion evidence; the METR and Redwood publications describe substantially the same independent investigation of supplied records. They should not be counted as independent replications merely because they appear on different sites.

We downloaded the public `full-wiki-logs.zip` export on September 5, 2026 and verified all five expanded-file checksums against the download listing. Section 4.2 reports our temporal aggregation of its records. The manifest’s May 1 write-date cut applies to stored revisions; the probe events come from separate May and June DSE request logs. This is a collector-built release, not direct access to the original server logs. Appendix A distinguishes the export totals from the larger narrative and explorer views.

The retained revisions comprise 13,403 from DSE, 1,013 from Probier, 169 from Fractal, and six from Dorfwiki. We matched 10,518 save events on 3,898 page keys to a later deletion event on the same key, confirming that the export retains such histories. These counts do not establish complete recovery of deleted content. We use event times and page keys for this check, not current page survival.

The collectors define probes as a narrow category of DSE requests attempting executable script injection. The 101 records are request-log based, and none records observed success. They are neither all exploratory requests nor confirmed successful exploits. Their earliest timestamp is May 17; no authenticated run identity connects these requests to subsequent saves. The event ordering alone cannot establish a formation sequence.

The working kit includes `analysis/wiki/analyse.py`, aggregate daily counts, and manifests for source acquisition and analysis results. The script accepts only the pinned export; acquisition and analysis-run dates are recorded separately. It rebuilds Figure 3 by parsing downloaded text without executing it. Nonempty labels are counted verbatim, without merging aliases or identifying people. No raw incident dataset, individual messages, or address records are redistributed with the manuscript.

Research citations identify the versions examined where results depend on a particular preprint. Cited experiments have not been reproduced; our new analysis is limited to the public export. The control-evaluation protocol remains prospective. The source archive contains the manuscript, editable schematic figures and tables, the rendered data figure, bibliography, and compiled bibliography. The working kit adds the analysis script and aggregate outputs; review and revision history remain separate repository records.

G Conventions for describing evidence

Evidence is described along separate dimensions rather than assigned a single grade. Its *basis* may be an incident artifact, an experiment, a formal argument, a standard, or an author proposal. Its *verification status* may be operator-reported, independently analysed from supplied data, independently reproduced, or unresolved. A primary source can still be self-reported, and an independent analysis can still have restricted access.

The manuscript uses ordinary attribution and citations to express these distinctions at the relevant claim. “Reported” identifies a source’s finding; “proposed” identifies an unvalidated control or test. Missing observations remain unknown unless the collection and search method support an absence claim. Numerical illustrations and conceptual diagrams are identified as such and should not be read as experimental results.

Table 1 gives the policy and observation questions used in Section 2.

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